

A Coarse-to-Fine Multimodal Detection Framework Based on Deep Learning for Robotic Coating Tasks

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Abstract—Automated robotic systems are evolving rapidly in industrial manufacturing. In robotic seal coating scenarios, while RGB-D based perception is adaptable for flexible demands, it struggles with achieving high accuracy. To address this limitation, we propose a multimodal visual detection system for robotic seam gluing, which employs a coarse-to-fine strategy across two- and three-dimensional (2-D and 3-D) data modalities. The proposed framework integrates two main components: a 2-D seam segmentation & classification network for coarse seam identification at the image level, and a 3-D seam refinement & transformation network (3D-SRTN) for precise seam refinement at the point cloud level. Specifically, 3D-SRTN performs local seam point optimization, followed with global transformation to minimize cumulative coordinate errors. A postprocessing procedure is then designed to generate smooth 6-DoF paths. Additionally, we introduce the RGB-D coating dataset, specifically tailored for this task, which includes both 2-D and 3-D annotated data for RGB-D coating perception. Ablation studies validate the effectiveness of individual modules, while comparative experiments demonstrate the overall superiority of our system.

Index Terms—Deep learning, path planning, RGB-D perception, seam extraction.

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I. INTRODUCTION

ROBOTIC seam seal coating is increasingly crucial in diverse industries, including aerospace, automotive, and construction. Traditional teach-playback methods struggle to adapt to the demands of modern flexible manufacturing [1], leading to the widespread adoption of vision-based perception for robotic tasks [2], [3]. Precise target recognition and localization are paramount for successful automation [4], [5], [6]. While laser vision systems offer high accuracy, RGB-D sensors provide a compelling alternative due to their superior scanning efficiency and lower cost. However, RGB-D-based seam perception is susceptible to inaccuracies stemming from various sources throughout the processing pipeline, including seam segmentation errors, accumulated coordinate transformation errors, and post-processing inaccuracies. Therefore, improving the accuracy and reliability of RGB-D-based seam coating remains a significant research challenge.

Regarding industrial seam detection, extensive research has been conducted, broadly classified into passive and active strategies. Passive approaches primarily utilize two-dimensional (2-D) images captured by vision cameras to predict seam locations, employing conventional image processing techniques [7], [8] or deep learning-based methods [9], [10]. However, these methods are often inadequate for complex gluing scenarios, as the absence of 3-D information restricts their application in high-precision tasks.

For active sensing, two primary types of visual sensors are commonly used: laser vision sensors and RGB-D sensors. Methodologies based on laser vision sensors often employ on-line tracking strategies [11], [12]. Achieving real-time performance necessitates continuous acquisition of laser line images and the use of computationally efficient algorithms for rapid seam point identification. However, the inherent local search mode significantly limits data collection efficiency. Furthermore, their reliance on predefined approximate paths derived from computer aided design (CAD) models or teaching playback reduces adaptability in flexible manufacturing environments.

To this end, RGB-D cameras have emerged as valuable tools in industrial seam detection operations [13]. Utilizing coordinate attention mechanism, Li et al. [14] improved the YOLOv5 to enhance weld groove feature detection in cluttered backgrounds.

Zhao et al. [15] modified the YOLOv8s-seg to a lightweight model for fast reasoning. To overcome the environment disturbance, TransUNet was proposed by Eren et al. [16] to achieve robust seam segmentation. However, these studies mainly focused on the 2-D pixel-level prediction, lacking the comprehensive understanding at the 3-D point-level.

Point-based approaches traditionally rely on geometric features for target seams localization [17], [18], [19]. Liu et al. [20] introduced an enhanced K-means clustering algorithm, leveraging geometric information to achieve precise seam localization. Apart from purely 3-D methods, hybrid 2-D–3-D approaches can provide regions of interest for 3-D prediction. [21]. Wei et al. [22] integrated FastSAM for image segmentation with a region growing algorithm for seam edge detection. Pan et al. [23] combined the YOLO-CE model and a variant of RANSAC algorithm for efficient seam extraction. Nonetheless, these methods often suffer from complex extraction pipelines, low efficiency, the need for manual threshold tuning, and limited generalization capabilities.

Consequently, learning-based methods such as WeldNet [24] emerged, which introduced a voxel-based network to directly segment the seam points from 3-D point clouds. Fernández et al. [25] developed UNet $\mathcal{L}++$, an architecture based on UNet++, to learn seam features from voxelized point cloud input. However, their inherent sparse sampling characteristics make it difficult to achieve the fine seam point segmentation.

Beyond seam detection, calibration significantly influences the system performance [26]. To address calibration errors of Denavit–Hartenberg parameters, Kong et al. [27] proposed two neural networks to predict compensation values for static and dynamic parameters in real-time. Utilizing the 3-D artifact and trigger probe, Fan et al. [28] employed the sand cat swarm optimization algorithm to estimate the kinematic parameter errors. For hand-eye calibration, Chen et al. [29] proposed a novel dual quaternion optimization method that separately minimizes rotational and translational errors. Additionally, actor-critic network is adopted by Li et al. [30] to optimize the parameters. Despite these advancements, individual calibration of parameters within the visual system can still lead to cumulative errors during matrix multiplication, even when each step is accurately calibrated. Furthermore, some transformations—such as camera intrinsic parameters and the transformation between color and depth cameras—are often set based on factory defaults without additional calibration, which can further compromise system accuracy.

To address the problems previously mentioned, we propose a learning-based multimodal framework for the coating seam detection and robotic trajectory generation Fig. 1. The main contributions of this study are as follows.

- 1) *2-D–3-D fusion neural network pipeline*: To tackle the inefficiency and limited generalization of traditional 3-D methods, we propose a novel framework combining *2-D seam segmentation & classification network (2D-SSCN)* for pixel-level coarse extraction and *3-D seam refinement & transformation network (3D-SRTN)* for point-level finetuning, followed by 6-D trajectory postprocessing for high-precision seam tracking.

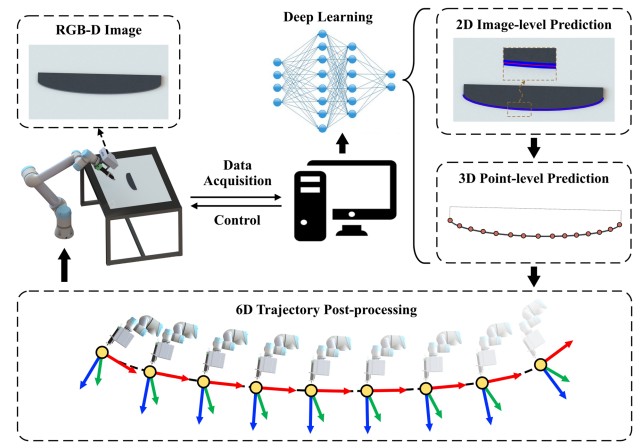


Fig. 1. Framework of the proposed multimodal planning method. This system utilizes an RGB-D camera for perception, and feeds the RGB-D image into the deep learning architecture for seam extraction. In this neural network, the 2-D image-level prediction coarsely segment the seam region in thick and thin shapes, while the 3-D point-level prediction refines the points to more appropriate locations. Ultimately, the post-processing generates the smooth 6-DOF trajectory.

- 2) *Seam feature refinement module (SFRM)*: To address the sparse sampling of 3-D learning-based algorithms and better utilize the seam points geometry, our SFRM within 3D-SRTN leverages the region of interest (RoI) from 2D-SSCN to derive coarse seam points, then guides them toward regions with prominent seam features for enhanced accuracy.
- 3) *Adaptive points transformation module (APTm)*: To mitigate cumulative system errors, our APTm learns an end-to-end transformation matrix within 3D-SRTN, enabling direct global correction of seam locations for improved robustness.
- 4) *RGB-D coating dataset*: We introduce a first-of-its-kind RGB-D Coating Dataset with rich 2-D and 3-D annotations, facilitating advanced research in seam detection and robotic trajectory planning.

The rest of this article is structured as follows. Section II and Section III introduce the 2D-SSCN and 3D-SRTN, respectively. Then, the 6-D trajectory postprocessing is described in Section IV. We present our RGB-D coating dataset in Section V. The experiments are conducted to evaluate the performance of the proposed method in Section VI. Ultimately, we Section VII concludes this article.

II. 2-D SEGMENTATION AND CLASSIFICATION

As illustrated in the top left of Fig. 2, the architecture of 2D-SSCN comprises two U-Net models followed by a classification head. The RGB-D image serves as the input to the cascaded U-Net models, which segment it into a thick mask and a thin mask with thicknesses of ξ_a and ξ_b pixels, respectively. These two masks are then concatenated and fed into the classification head to predict the seam type.

In 3D-SRTN, the thick mask serves as the RoI for feature extraction, while the thin mask provides the seam points for

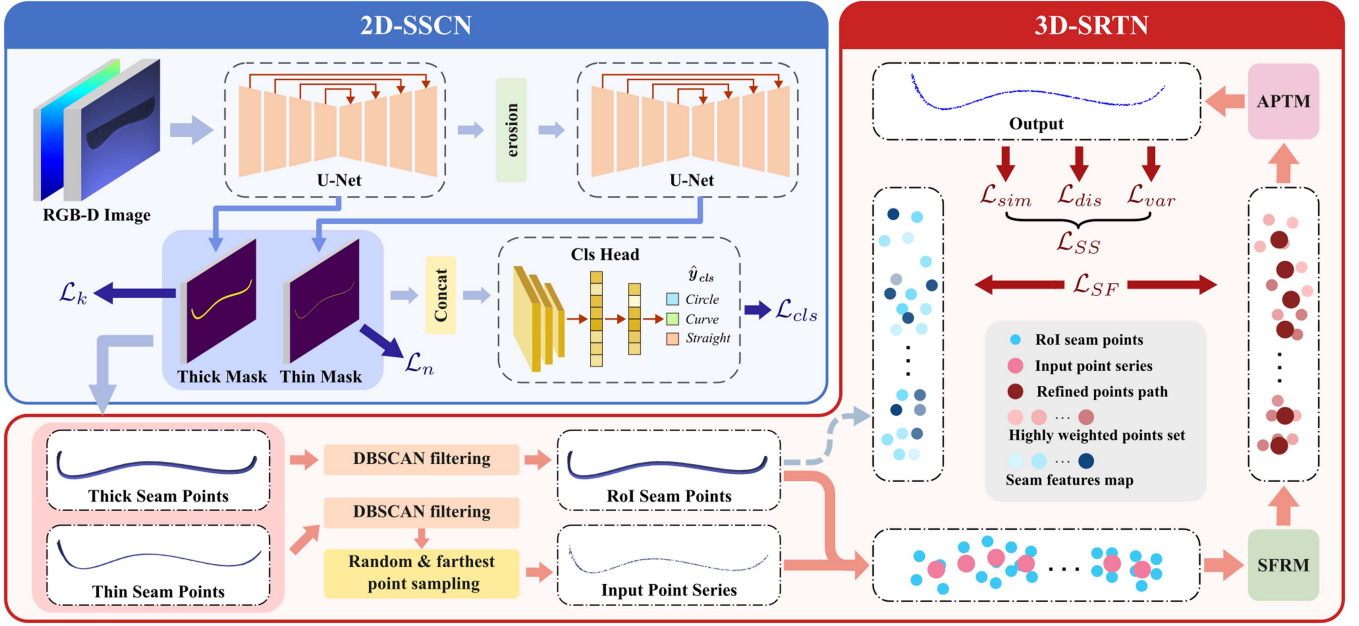


Fig. 2. Architecture illustration of our coarse-to-fine deep learning model, which is composed of two essential components: 2D-SSCN and 3D-SRTN. In 2D-SSCN (Section II), RGB-D image is fed to a cascade of two U-Net, deriving the thick mask and thin mask of coating seam, then the class label is obtained by Cls Head. Then in 3D-SRTN (Section III), two thickness of masks are reconstructed into 3-D point cloud, and pre-processed as the input of network. After passed through SFRM and APTM, we can derive the refined and transformed point path.

optimization. To ensure robustness, ξ_a must be sufficiently large (≥ 10 pixels), whereas ξ_b should be minimized (ideally 1–5 pixels). Given these constraints, high segmentation accuracy can be achieved for the thick mask. However, thin mask prediction may suffer due to ξ_b being comparable to the convolution kernel size, complicating the network’s ability to distinguish seams from the background.

To this end, we utilize the precise result of the thick mask to generate the coarse thin mask. A sliding convolution kernel erodes the thick mask, and the result is refined by the subsequent U-Net to produce the thin mask. Among kernel shapes (rectangular, elliptical, cross-shaped), the cross-shaped kernel exhibits weaker erosion but superior sensitivity to high-gradient pixels. This property preserves skeleton integrity and enhances corner-point detection, making it ideal for line-shaped image processing.

The classification head begins with a convolutional encoder to down-sample feature maps, followed by flattening into embeddings. A multilayer perceptron (MLP) then processes these embeddings to predict the class label.

2D-SSCN adopts a two-stage training strategy:

- 1) *Thick-seg stage*: The initial U-Net predicts the thick mask, optimized via binary cross-entropy (BCE) loss:

$$\mathcal{L}_{\text{thick}} = \text{BCE}(\hat{M}_k, M_k), \quad (1)$$

where $\hat{M}_k$ and M_k denote the predicted and ground-truth thick masks, respectively.

- 2) *Thin-seg stage*: After thick-seg stage converges, the first U-Net’s weights are frozen. The second U-Net predicts the thin mask, while the classification head predicts the

class label, jointly optimized via a multitask loss:

$$\mathcal{L}_{\text{thin}} = \text{BCE}(\hat{M}_n, M_n) + \text{CE}(\hat{y}_{\text{cls}}, y_{\text{cls}}), \quad (2)$$

where $\hat{M}_n$ and $\hat{y}_{\text{cls}}$ are the predicted thin mask and class label respectively, M_n and y_{cls} are their ground truths, and $\text{CE}(\cdot)$ represents the multiclass cross-entropy (CE) loss.

Importantly, decoupling segmentation and classification enables per-class mask generation without interclass competition. The dedicated classification branch improves segmentation accuracy by avoiding label conflicts [31].

III. 3-D REFINEMENT AND TRANSFORMATION

A. Seam Points Preprocessing

As shown in the bottom left of Fig. 2, a preprocessing pipeline is applied to convert 2-D segmentation results into 3-D representations for downstream processing. Initially, the two 2-D predicted masks from 2D-SSCN are mapped onto 3-D point clouds with the aligned depth map, thereby generating thick and thin seam points.

However, mis-segmentation and potential capture failures, outlier points are commonly observed in the reconstructed point clouds. To mitigate this, we employ density-based spatial clustering of applications with noise (DBSCAN) clustering to filter noise by adaptively grouping points into clusters and removing sparse clusters (those with fewer points than a predefined threshold). The remaining thick seam points are then designated as the RoI seam points, denoted as P_{RoI} .

To obtain fixed-size inputs for training from the filtered thin seam points, a sampling strategy is employed. Both farthest point

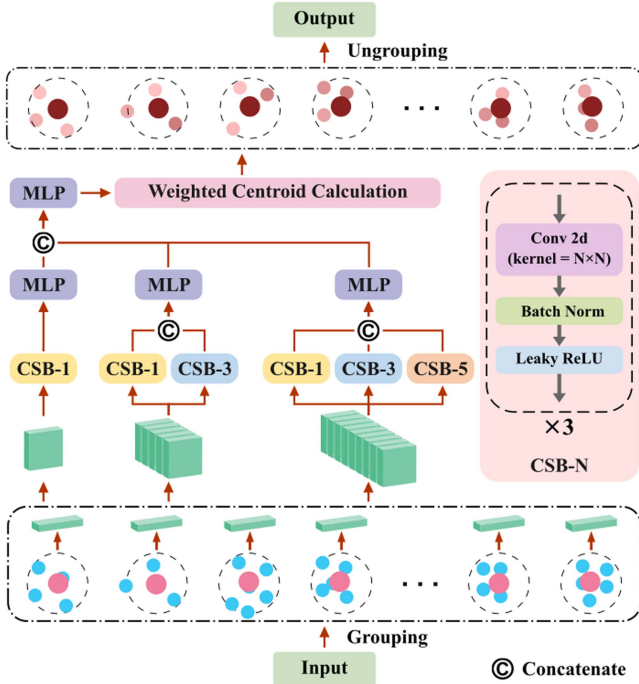


Fig. 3. Architecture of the seam feature refinement module (SFRM).

sampling [32] and random point sampling are used. Farthest point sampling selects a small subset of points (approximately 5% of the final sample size N_I) to ensure inclusion of edge and end points. The remaining points are then randomly sampled to guarantee a uniform distribution along the sealing path, deriving the input consecutive point series S_I .

B. Seam Feature Refinement Module (SFRM)

The main idea of SFRM is to optimize each single point in S_I , reducing errors caused by inaccurate annotations, improper segmentation, misalignments between color and depth maps, and so forth.

As illustrated in Fig. 3, for each point $p_i S_I$, we first query K_n neighbors from P_{RoI} , grouped as the i th point group $x_i \mathbb{R}^{C \times K_n}$ (C denotes the channel size). To further leverage the feature relationships among adjacent points, a hierarchical multiscale feature embedding strategy is employed. Specifically, for each x_i , we gather point groups from its bidirectional neighbors $\{x_{i-k}, x_{i-k+1}, \dots, x_i, \dots, x_{i+k-1}, x_{i+k}\}$, and stack them to form the grouped points set g_i . Subsequently, similar step is taken to stack the adjacent g_i into feature maps $\{X_t \mathbb{R}^{N \times C \times (2k+1) \times K_n} \mid t = 0, 1, 2\}$ at three scales ($k = 0, 2, 4$).

The encoder employs a progressive multiscale convolution approach. We implement a convolutional sequence block (CSB- N) architecture with varying kernel sizes ($N = 1, 3, 5$), containing three sequential sub-blocks: a convolutional layer, batch normalization, and LeakyReLU activation. Initially, X_1 is processed through CSB-1 module. X_2 is concurrently fed into both CSB-1 and CSB-3 module, with their outputs concatenated. Similarly, X_3 is processed through CSB-1, CSB-3, and CSB-5 module

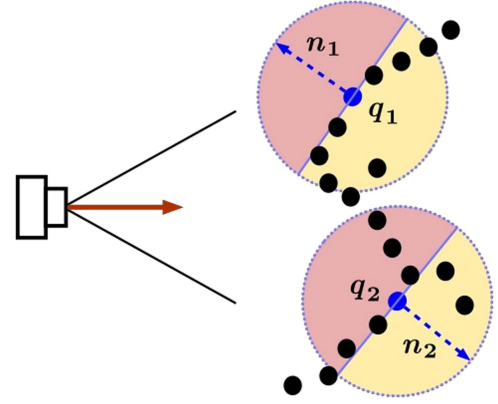


Fig. 4. Conceptual representation of the concavity feature. n_1 indicates q_1 's dominant normal vector which point towards the side with fewer neighbors. Within each neighborhood sphere, the tangent plane divides the sphere into two regions: the red region, closer to the camera, includes concave points; the yellow region, farther away from the camera, involves convex points.

simultaneously. Separately projected through MLP networks, the resulting features are concatenated, and processed by a MLP to generate the probability map $G_p \mathbb{R}^{N_I \times K_n}$ (N_I is the point number of S_I) via LogSoftmax.

By selecting K_w points with highest weights from each neighboring group x_i according to G_p , highly weighted points set $Y_w \mathbb{R}^{N \times 3 \times K_w}$ are obtained, where three channels indicate 3-D coordinates (x, y, z). The refined point path $S_r \mathbb{R}^{N \times 3}$ is then computed as the weighted average of Y_w .

C. Seam Feature Point-Wise Loss

Now, we introduce the proposed novel seam feature point-wise loss, $\mathcal{L}_{SF}$. The core idea behind this loss function is to assign higher probabilities in G_p to points with distinctive seam features. Typically, points near seams exhibit higher curvature and concavity from the camera's view. Concavity suggests that the neighbors of a point protrude outward along the camera's optical axis.

The core principle of SFRM lies in learning an implicit representation of seam points from explicit geometric features. Accordingly, we design the concavity and curvature feature values for loss calculation. For each point $q_t P_{RoI}$, we define its curvature and concavity feature values as $K_t K$ and $C_t C$, respectively, where K and C represent the curvature and concavity feature maps.

As shown in Fig. 4, we query q_t 's N_n neighboring points Q_t^n within a neighborhood sphere. The dominant normal vector n_t guides the pointing direction of the glue gun. It is evident that points whose normals align closely with the camera's Z-axis are highly likely to be seam points.

To quantitatively describe this, the concavity feature value is designed as $C_t = \frac{N_{conc}}{N_n}$, where N_{conc} represents the number of concave points. As shown in Fig. 4, C_2 is much larger than C_1 , indicating that p_2 is more likely to be a seam point than p_1 .

Following that, chord and normal vectors method [33] is used to estimate the curvature κ_t . The curvature feature value K_t is

then obtained through global and local normalization followed by averaging, as demonstrated in the following:

$$\hat{\kappa}_t = \max(\sqrt{\kappa_t}, t_c), \quad (3)$$

$$\hat{\kappa}_t^g = N_{\text{global}}(\hat{\kappa}_t) = \frac{\hat{\kappa}_t - \hat{\kappa}_{\min}}{\hat{\kappa}_{\max} - \hat{\kappa}_{\min}}, \quad (4)$$

$$\hat{\kappa}_t^l = N_{\text{local}}(\hat{\kappa}_t^g) = \frac{\hat{\kappa}_t^g - \hat{\kappa}_{\min}^g}{\hat{\kappa}_{\max}^g - \hat{\kappa}_{\min}^g}, \quad (5)$$

$$K_t = \frac{1}{N_n} \sum_{i \in I_t^n} \hat{\kappa}_i^l, \quad (6)$$

where t_c is the threshold to clip excessively high values; and $\hat{\kappa}_t^g$ and $\hat{\kappa}_t^l$ indicate the globalized and localized curvature, respectively. Here, $N_{\text{global}}(\cdot)$ performs the global min-max normalization, with $\hat{\kappa}_{\min}$ and $\hat{\kappa}_{\max}$ being the minimum and maximum values of $\hat{\kappa}_t$ across points in $\mathbf{P}_{\text{RoI}}$. Similarly, $N_{\text{local}}(\cdot)$ applies local min-max normalization, where $\hat{\kappa}_{\min}^g$ and $\hat{\kappa}_{\max}^g$ is the minimum and maximum values of $\hat{\kappa}_t^g$ for points in Q_t^n . The indices of Q_t^n are denoted by I_t^n .

After calculating the concavity and curvature feature maps $\mathbf{C}$ and $\mathbf{K}$, we extract the feature value of each point in $\mathbf{Y}_w$ from the point-wise combined seam feature map $\mathbf{F}^p$, defined as follows:

$$\mathbf{F}^p = \mathbf{K} + \alpha \mathbf{C}, \quad (7)$$

where α is a weighting parameter.

The seam feature point-wise loss, $\mathcal{L}_{\text{SF}}$, is formulated as follows:

$$\mathcal{L}_{\text{SF}} = \frac{1}{M_w} \sum_{i=1}^{M_w} \exp \left(-\frac{1}{N_I} \sum_{j=1}^{N_I} G_{ij} F_{ij}^p \right), \quad (8)$$

where p^{ij} denotes the i th neighbor point in $\mathbf{Y}_w$ corresponding to the j th path point in $\mathbf{S}_I$. The combined feature value for p^{ij} is indicated as F_{ij}^p , and its feature weight can be queried from $\mathbf{G}_p$ as G_{ij} . By calculating the weighted feature values of points in $\mathbf{Y}_w$ for back-propagation, $\mathbf{Y}_w$ can converge towards points with higher feature values, improving the prediction of $\mathbf{S}_r$.

D. Adaptive Points Transformation Module (APTM)

The main consideration of APTM is to eliminate the error cumulated from transformations of coordinates, caused from eye-on-hand calibration, depth camera calibration, robot arm kinematic parameters calibration, end-effector assembly, and similar processes. To address this, APTM generates a transformation matrix to refine $\mathbf{S}_r$ to a more accurate location.

As illustrated in Fig. 5, we predict an affine transformation matrix by a T-net [34] and apply it to $\mathbf{S}_r$. Then, three 1×1 Conv-ReLU layers encode the input into hidden feature. After max pooling function, we flatten the hidden feature into embedding vector, which is decoded by MLPs into six tokens $t_d \in \mathbb{R}^6$, three for translations (x, y, z) and three for rotations (roll, pitch, yaw). To promote faster convergence, we employ the tanh activation to limit the value of t_d in the error bound priors $\mathbf{b} \in \mathbb{R}^6$, which can be formulated as follows:

$$\mathbf{t}_b = \tanh(\mathbf{t}_d) * \mathbf{b} \in \mathbb{R}^6 \quad (9)$$

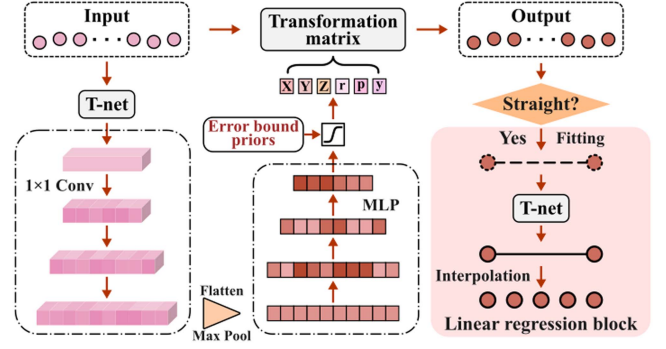


Fig. 5. Architecture of the adaptive points transformation module (APTM).

where $\mathbf{t}_b$ is the final transformation in the real-world, and for $\mathbf{b}$, we set the error bounds to ± 5 mm for translation and $\pm 10^\circ$ for rotation. Ultimately, adjustment transformation matrix generated from $\mathbf{t}_b$ transforms $\mathbf{S}_r$ to the transformed points path $\mathbf{S}_t \in \mathbb{R}^{N \times 3}$.

In particular, the path can be represented by two key points and the line connecting them. To extract these points, we additionally introduce a linear regression block. First, principal component analysis estimates the principal direction of $\mathbf{S}_t$. By projecting $\mathbf{S}_t$ onto this direction, two endpoints can be subsequently identified to define the line equation. Afterward, T-nets generate two transformation matrices to finetune the positions of these endpoints. Finally, equidistant interpolation along the adjusted line yields the transformed points path.

E. Seam Structure Point-Wise Loss

Based on the seam structure, we propose the point-level loss $\mathcal{L}_{\text{SS}}$ that considers similarity, distance, and distribution.

First, we employ the Chamfer distance to measure the structural similarity between $\mathbf{S}_t$ and the ground truth path $\mathbf{S}_{gt}$. For each point $p \in \mathbf{S}_t$, we derive its nearest point $q \in \mathbf{S}_{gt}$, and vice versa for points in $\mathbf{S}_{gt}$. The similarity loss $\mathcal{L}_{\text{sim}}$ is defined as the sum of bidirectional average distances between $\mathbf{S}_t$ and $\mathbf{S}_{gt}$

$$\mathcal{L}_{\text{sim}} = \frac{\sum_{p \in \mathbf{S}_t} \min_{q \in \mathbf{S}_p} \|p - q\|_2^2}{|\mathbf{S}_t|} + \frac{\sum_{q \in \mathbf{S}_p} \min_{p \in \mathbf{S}_t} \|p - q\|_2^2}{|\mathbf{S}_p|}. \quad (10)$$

Then, we utilize the root mean square error (RMSE) loss between $\mathbf{S}_t$ and $\mathbf{S}_{gt}$ as the distance loss $\mathcal{L}_{\text{dis}}$

$$\mathcal{L}_{\text{dis}} = \sqrt{\frac{1}{|\mathbf{S}_t|} \sum_{p \in \mathbf{S}_t} \min_{q \in \mathbf{S}_p} \|p - q\|_2^2}. \quad (11)$$

Additionally, to ensure the uniform distribution of $\mathbf{S}_t$, the variance of the distances between adjacent points is taken as the variance loss $\mathcal{L}_{\text{var}}$, expressed as follows:

$$\mathcal{L}_{\text{var}} = \text{Var} \left(\{ \|p_{i+1} - p_i\| \}_{i=1}^{N-1} \right). \quad (12)$$

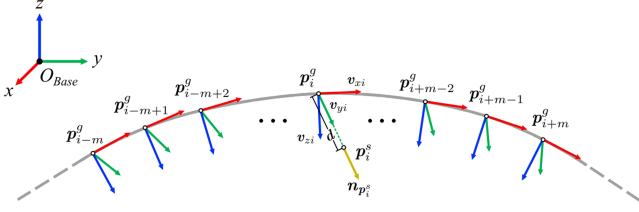


Fig. 6. Determination of the 6-DOF translation and rotation for each sealing seam point.

So, the seam structure point-level loss $\mathcal{L}_{SS}$ can be calculated by the following:

$$\mathcal{L}_{SS} = \lambda_s \mathcal{L}_{sim} + \lambda_d \mathcal{L}_{dis} + \lambda_v \mathcal{L}_{var}, \quad (13)$$

where λ_s , λ_d , and λ_v are weight parameters.

As illustrated in Fig. 2, we train SFRM and APTM in one-stage, so the overall loss function for 3D-SRTN can be formulated as follows:

$$\mathcal{L} = \mathcal{L}_{SF} + \mathcal{L}_{SS}. \quad (14)$$

IV. 6-D TRAJECTORY POSTPROCESSING

In the 6-D trajectory postprocessing, we initially apply a uniform process to all types of pieces. First, we register the point clouds of workpieces captured from multiple views with the point-to-plane iterative closest point method. Afterward, we extract the skeleton of registered points path using the Laplacian-based contraction (LBC) [35] algorithm. This algorithm can contract the point cloud while preserving the overall structure through localized geometric processing.

Next, varied fitting operations are adopted to refine diverse kinds of seams. For linear and circular configurations, we use least squares regression analysis, to accurately model the 3-D linear and circular equations. As for the curve-type workpieces, cubic spline curve fitting is adopted to generate smooth spline equations. These equations are uniformly interpolated at equal distances to produce a sequence of points $\{p_1^s, p_2^s, \dots, p_{n_f}^s\}$. The number of interpolated points, n_f , depends on the specific application needs.

As shown in Fig. 6, to guide the gluing process, each interpolated seam point p_i^s is refined as the glue path point p_i^g , using the relation: $p_i^g = p_i^s - d * n_{p_i^s}$, where d is the optimal distance to glue a workpiece suggested by different sets of gluing equipment, and $n_{p_i^s}$ is the dominant normal vector at p_i^s . Having determined the 3-DOF translation parameters $\{p_1^g, p_2^g, \dots, p_{n_f}^g\}$, 3-DOF rotation parameters $\{v_1, v_2, \dots, v_{n_f}\}$ are also required to fully define the gluing trajectory. Each v_i comprises three orthogonal unit vectors $(v_{xi}, v_{yi}, v_{zi})^T$. v_{xi} represents the moving direction of the glue gun, defined as the direction from p_i^g to p_{i+1}^g . v_{zi} represents the pointing vector of the glue gun, defined as the dominant normal vector of p_i^g . To reduce disturbance, v_{xi} and v_{zi} are defined as the average of $(2m+1)$ nearest vectors, i.e., $v_{xi} = (\sum_{j=i-m}^{i+m} v_{xj}) / (2m+1)$, $v_{zi} = (\sum_{j=i-m}^{i+m} v_{zj}) / (2m+1)$. v_{yi} is then computed via the cross product $v_{yi} = v_{zi} \times v_{xi}$, in accordance with the right-hand rule. Consequently, each path

point has its own coordinate formed by $(v_{xi}, v_{yi}, v_{zi})^T$, thus a dense and evenly distributed 6-DOF trajectory for sealing is planned.

V. DATASET

Currently, there is no existing dataset that addresses the seam gluing task at both image and point-cloud levels using RGB-D cameras. To facilitate evaluation, we have created a novel dataset called the RGB-D Coating Dataset, which comprises two subsets: the 2D-SSCN subset and the 3D-SRTN subset, corresponding to the two main components of our approach. During data collection, we capture both 2-D and 3-D information from 13 workpieces, including 4 straight-shaped parts, 5 curved-shaped parts, and 4 circular-shaped parts. In total, 1690 data groups are collected, covering various positions and viewpoints, with 130 groups per workpiece.

A. 2D-SSCN Subset

Each group consists of RGB image $I_{RGB} \in \mathbb{R}^{H \times W}$, depth image $I_D \in \mathbb{R}^{H \times W}$, workpiece type label $y_{cls} \in \mathbb{R}$, thick mask ground truth $M_k \in \mathbb{R}^{H \times W}$ and thin mask ground truth $M_n \in \mathbb{R}^{H \times W}$, where H and W represent the height and width of RGB image, respectively. I_{RGB} s and I_D s can be obtained by capturing components from multiple perspectives with RGB-D camera. Furthermore, for y_{cls} , we labeled each image among three workpiece types: straight, curve, and circle. LabelMe is employed to annotate the target seam as line strips in RGB image, creating a sequence of ordered pixel points, then fitted to a skeleton. Subsequently, we can derive M_k and M_n by tracing the skeleton with a brush of thickness ξ_a and ξ_b pixels, respectively, with OpenCV.

B. 3D-SRTN Subset

Each group consists of 3-D ground truth path $S_{gt} \in \mathbb{R}^{N_{gt} \times 3}$ and the homogeneous transformation matrix ${}^B T_C \in \mathbb{R}^{4 \times 4}$ from robot base to camera, where N_{gt} indicates the number of path points. For S_{gt} s, a series of gluing waypoints is set by an experienced gluer in the teach-playback method, then linear and circular equations are selected to estimate the seams of straight-type and circle-type workpieces, respectively, interpolating them into N waypoints in average distributions. Meanwhile, for curve-type workpieces, spline interpolation is adopted to evenly distribute the numbers of waypoints to N . With regard to ${}^B T_C$, for each capture pose, the transformation from the robot end to the camera, ${}^E T_C$, can be derived through eye-on-hand calibration, while the transformation from the robot base to the robot end, ${}^B T_E$, resulted from forward kinematics. Ultimately, the base to camera transformation ${}^B T_C$ can be obtained by ${}^B T_C = {}^B T_E {}^E T_C$.

VI. EXPERIMENTS

In this section, we first present the training details of our proposed framework (Section VI-A). Subsequently, the prediction results are showcased in Section VI-B. To validate the effectiveness of the individual modules, ablation studies are conducted

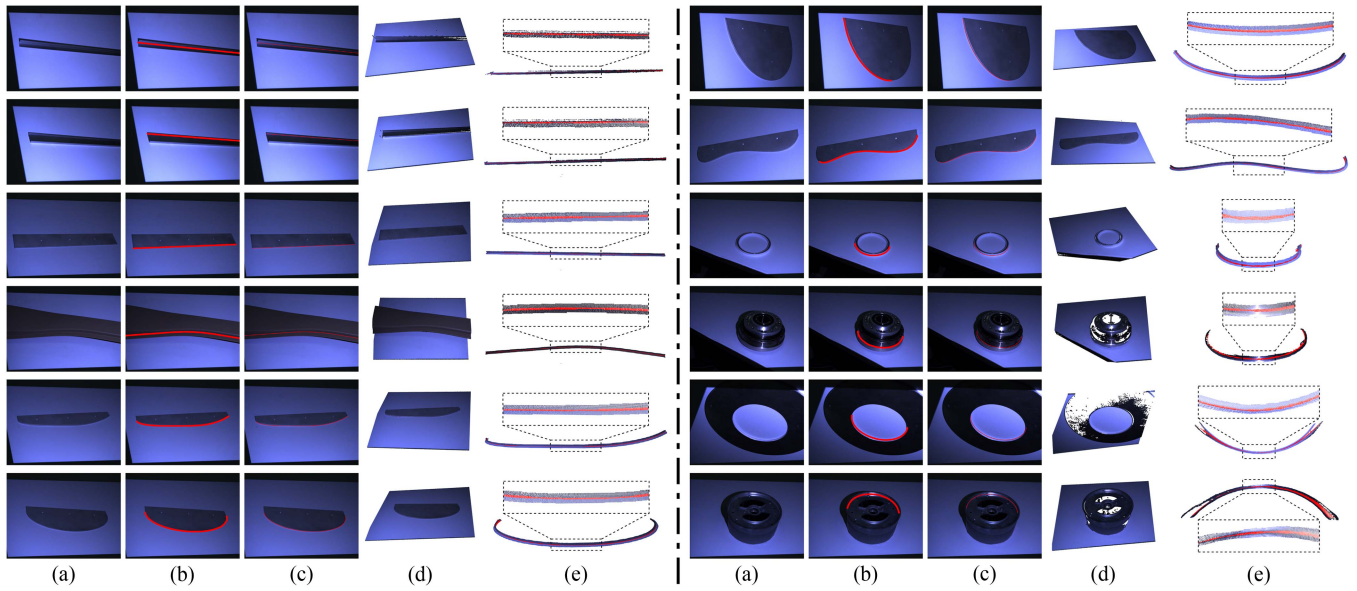


Fig. 7. Segmentation results of 2D-SSCN. (a) RGB images. (b) Thick masks. (c) Thin masks. (d) Reconstructed scenario point clouds. (e) Thick seam points (original color) and thin seam points (red color).

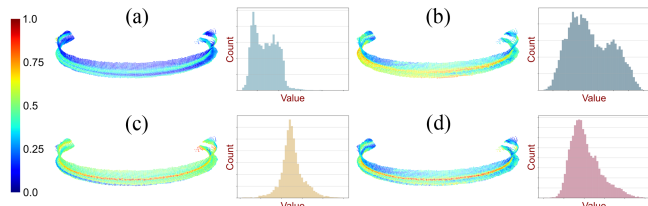


Fig. 8. Visualization of the seam feature extraction results. (a)–(d) Heatmaps of globalized curvature, localized curvature, concavity and combined seam feature, along with their corresponding distribution histograms.

TABLE I
MODEL PERFORMANCE OF 2D-SSCN

Metrics		Value (%)		
		Straight	Curve	Circle
IoU	thick mask	97.27	96.94	95.49
	thin mask	91.09	89.98	86.05
Accuracy	thick mask	99.90	99.89	99.90
	thin mask	99.93	99.93	99.93
	classification	99.75	99.90	100.00

(Section VI-C). Additionally, to better demonstrate the superiority of our approach, we perform comprehensive comparison experiments with various existing methods (Section VI-D). Finally, we apply the framework to a real-world physical scenario to evaluate its practical performance (Section VI-E).

A. Training Details

All experiments were performed using Python and the PyTorch framework. The computational infrastructure comprised a master workstation equipped with an Intel Core i7-12700KF

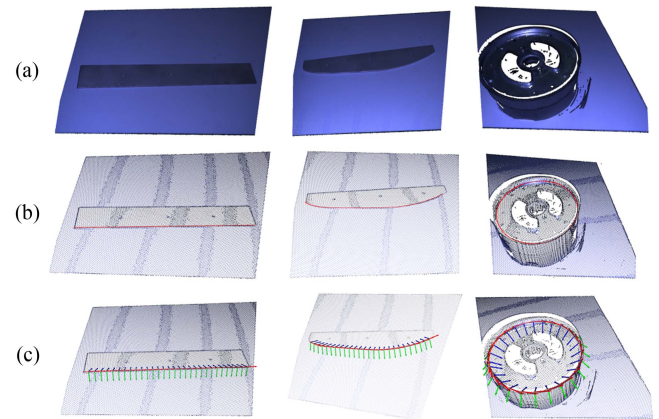


Fig. 9. Results of 6-D trajectory postprocessing. (a) Reconstructed point cloud. (b) Postprocessed 3-D path. (c) Postprocessed 6-D trajectory.

processor (5.0 GHz), 32 GB of RAM, and an NVIDIA RTX 3080 GPU.

To preserve image quality during processing, the original image size of 1080×1440 was maintained as the input resolution. The AdamW optimizer was employed, with the batch size, learning rate, and weight decay parameters set to 1, 0.001, and 0.0001, respectively. The key parameters used throughout the training pipeline are summarized as follows: $\xi_a = 20$, $\xi_b = 3$, $N_{gt} = 4096$, $N_I = 512$, $K_n = 128$, $K_w = 8$, $\alpha = 0.1$, $\lambda_s = 1$, $\lambda_d = 10$, $\lambda_v = 0.002$.

To ensure the robustness and generalizability of the proposed model, the RGB-D Coating Dataset, consisting of 1690 labeled data groups, was partitioned into training, validation, and testing sets in a 6:2:2 ratio. This resulted in 1014 data groups for training. Our model uses a three-stage training procedure: 2D-SSCN

TABLE II
ABLATION STUDY RESULTS

SFRM	APTM	Post-proc.	RMSE(mm)					MAE(mm)				
			Straight↓	Curve↓	Circle↓	Average↓	Gain	Straight↓	Curve↓	Circle↓	Average↓	Gain
✓	✓	✓	2.383	2.232	2.196	2.270	(baseline)	2.301	2.138	2.134	2.191	(baseline)
			2.068	1.978	1.804	1.950	-0.320	1.934	1.788	1.631	1.784	-0.407
			1.127	1.458	1.520	1.368	-0.902	1.003	1.242	1.378	1.208	-0.983
✓	✓	✓	1.143	1.593	1.476	1.404	-0.866	1.044	1.462	1.341	1.282	-0.909
			0.800	1.225	1.420	1.148	-1.122	0.735	1.104	1.282	1.040	-1.151
			0.593	1.099	1.252	0.981	-1.289	0.565	1.020	1.164	0.916	-1.275

thick-seg (1.6 h, 50 epochs), 2D-SSCN thin-seg (3.43 h, 80 epochs), and 3D-SRTN (26.27 h, 75 epochs). The total training time is 31.29 h.

B. Seam Prediction Results

1) *Results of Image-Level Prediction*: The 2D-SSCN model segments two thickness of mask from the inputs, and classifies the workpiece. Intersection over union (IoU) and accuracy metrics are used to evaluate the segmentation and classification of 2D-SSCN. As shown in Table I, the results indicate that the proposed model performs well in classification and thick mask segmentation, though suboptimal for thin masks. However, Fig. 7 demonstrates that despite the lower IoU for thin masks, the segmented pixels retain sufficient spatial coherence to reconstruct the seam's general outline in the corresponding point cloud. Since subsequent processing stages will involve pixel sampling and positional refinement, an exceptionally high prediction performance for thin masks is unnecessary. Thus, the current segmentation quality is deemed adequate for downstream tasks.

2) *Seam Feature Extraction*: To enable the model to refine seam point positions towards regions exhibiting prominent seam features, we calculate the curvature and concavity features for each point. As depicted in Fig. 8, the globalized curvature heatmap effectively highlights regions of high curvature, represented by red lines or curves. Then, the localized curvature heatmap refines the distribution of these curvature values, promoting a more uniform spread across the heatmap. This refinement mitigates the impact of local extrema on the computation of global feature values. Moreover, high curvature regions exhibit higher feature values in the localized curvature heatmap.

However, certain convex points in the curvature heatmap may exhibit undesirably high feature values. To address this, the concavity heatmap provides complementary information, effectively attenuating the feature values of convex points. Consequently, the combined seam feature heatmaps clearly delineate points characterized by both high curvature and high concavity, providing a robust representation of the seam.

3) *Results of Point-Level Prediction and Planning*: As shown in Fig. 9(b), the postprocessed 3-D path exhibits a smooth and continuous trajectory, which is achieved through skeleton extraction, fitting, and interpolation. Building upon this, the postprocessed 6-D trajectory in Fig. 9(c) incorporates both position and orientation information, derived by analyzing the dominant normal vector at each point. This results in a coherent

6-D path that reflects the underlying geometry and directionality of the trajectory, ensuring smooth and controlled movement of the robot.

C. Ablation Study

1) *Quantitative Experiments*: In our method, the 2D-SSCN model segments masks from RGB-D images in 2D-level, and provide the workpieces type labels, these results are irreplaceable for the following processing steps. To this end, we only evaluate the improvement of SFRM, APTM and post-processing steps, and choose 2D-SSCN as the baseline.

In this study, we choose the RMSE and the mean absolute error (MAE) as the metrics for path accuracy evaluation.

The impact of our various models is summarized in Table II. Compared to the baseline, our comprehensive model demonstrates an improvement of 1.289 mm in RMSE and 1.275 mm in MAE. Furthermore, the use of SFRM, APTM, and postprocessing each contributes significantly to these enhancements. Ultimately, the integration of all components results in the optimal model configuration.

2) *Qualitative Experiments*: The complete pipeline of our framework can be summarized as: 2D-SSCN → SFRM → APTM → 6-D postprocessing. To provide a more intuitive understanding of the impact of each module, we extracted and qualitatively compared four sets of intermediate processed seam points throughout this pipeline. As shown in Fig. 10, we present results for workpieces with straight, curved, and circular shapes. The seam points are visualized as follows:

- 1) sampled points (light red, output of 2D-SSCN after sampling);
- 2) refined points (dark red, output of SFRM);
- 3) transformed points (light blue, output of APTM);
- 4) postprocessed points (dark blue, output of 6-D trajectory postprocessing).

Furthermore, the corresponding ground truth trajectory points are plotted in purple.

As illustrated in Fig. 10, our point-level strategy shows strong suitability for all three workpiece types. Specifically, SFRM drives sampled points toward regions with stronger seam features, yielding refined points with more precise locations and more contracted outlines. Then APTM globally transforms refined points into transformed points closer to the ground truth, mitigating cumulative errors. Finally, 6-D postprocessing further contracts and fits transformed points, generating smooth and

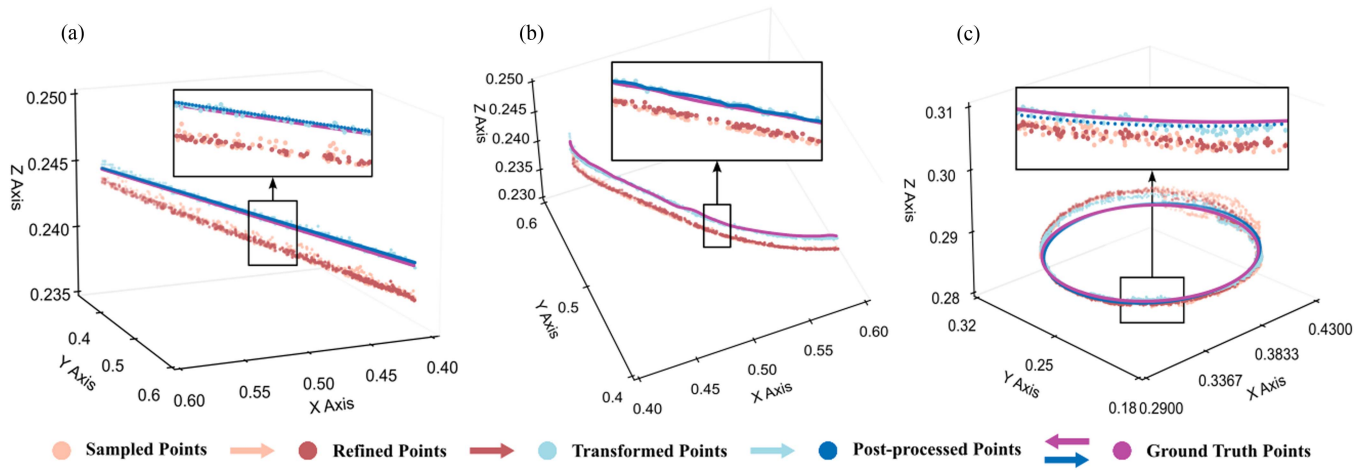


Fig. 10. Intermediate seam point processing results for (a) straight, (b) curved, and (c) circular workpieces.

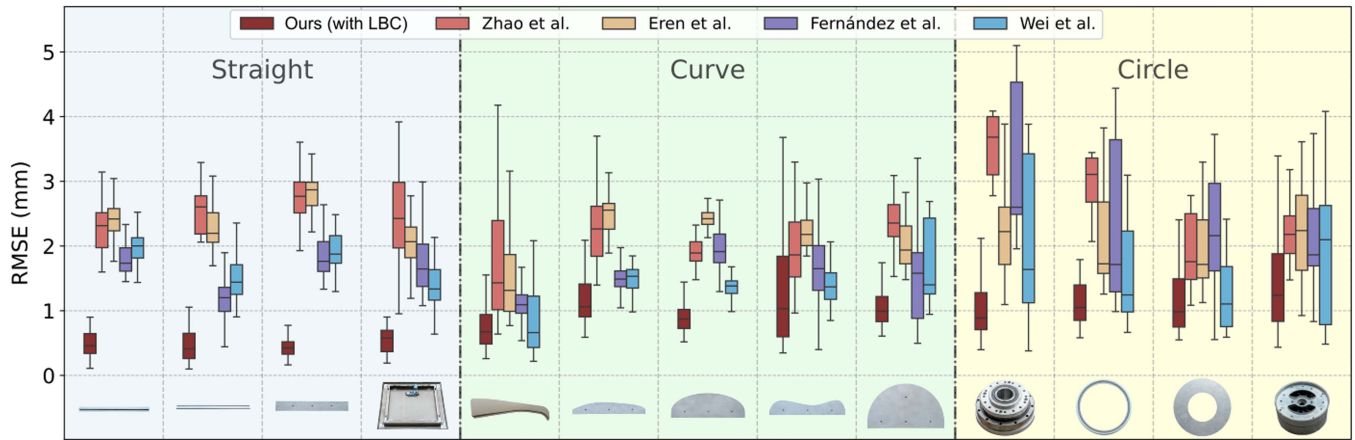


Fig. 11. RMSE comparison of prediction methods across workpieces.

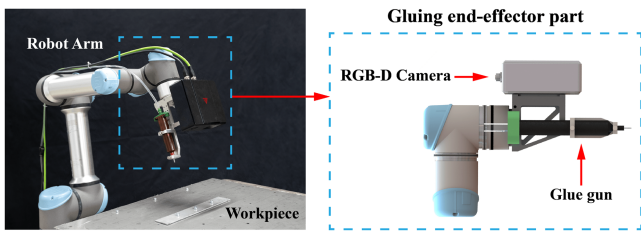


Fig. 12. (Left) Experimental gluing platform. (Right) End-effector.

evenly distributed post-processed points that align more closely with the ground truth.

D. Comparative Experiments

To validate the performance of our proposed method, we compare it against four representative techniques that also utilize RGB-D cameras for data acquisition:

- 1) Zhao et al. [15];
- 2) Eren et al. [16];
- 3) Fernández et al. [25];
- 4) Wei et al. [22].

Methods (1) and (2) are 2-D learning-based approaches: paper [15] employs a modified YOLOv8s-seg architecture, while paper [16] is based on the TransUNet model. Methods (3) and (4) are 3-D techniques: paper [25] predicts seams with voxel-based UNet $\mathcal{L}++$, and paper [22] first segments the RoI using FastSAM, then identifies seam points with traditional region-growing algorithm. Given the computational expense of the skeleton contraction method (LBC algorithm) in postprocessing, we evaluate our framework both with and without LBC to systematically quantify the accuracy-efficiency tradeoff.

As shown in Fig. 11 and Table III, our proposed system achieves relatively tighter interquartile range and lower RMSEs across three types of workpieces, indicating a marked improvement over existing methods documented in the literature. Specifically, methods (1) and (2) rely solely on 2-D image for seam segmentation, neglecting valuable spatial information in the seam geometry. Method (3), while utilizing point clouds, processes the entire captured scene with sparse sampling, failed in finegrained seam segmentation. Method (4), using predicted RoIs integrated with traditional 3-D seam extraction techniques, still suffers from limited adaptability to various workpiece geometries and ignorance of cumulative transformation errors. By

TABLE III
COMPARISON RESULTS

System	Method	Weight Size (M)	Time (ms)	RMSE (mm)			MAE (mm)		
				Straight	Curve	Circle	Straight	Curve	Circle
Zhao et al. [14]	2D Learning	4.4	357	2.496	2.062	2.486	2.407	1.882	2.197
Eren et al. [15]	2D Learning	8.4	408	2.389	2.172	2.132	2.343	2.012	2.052
Fernández et al. [24]	3D Learning	23.3	409	1.917	1.893	2.112	1.803	1.831	1.902
Wei et al. [21]	2D Learning + 3D Traditional	15.2	1219	1.644	1.388	1.631	1.584	1.301	1.363
Ours (w/o LBC)	2D Learning + 3D Learning	7.3	483	0.608	1.220	1.425	0.578	1.101	1.287
Ours (with LBC)	2D Learning + 3D Learning	7.3	693	0.593	1.099	1.252	0.565	1.020	1.164

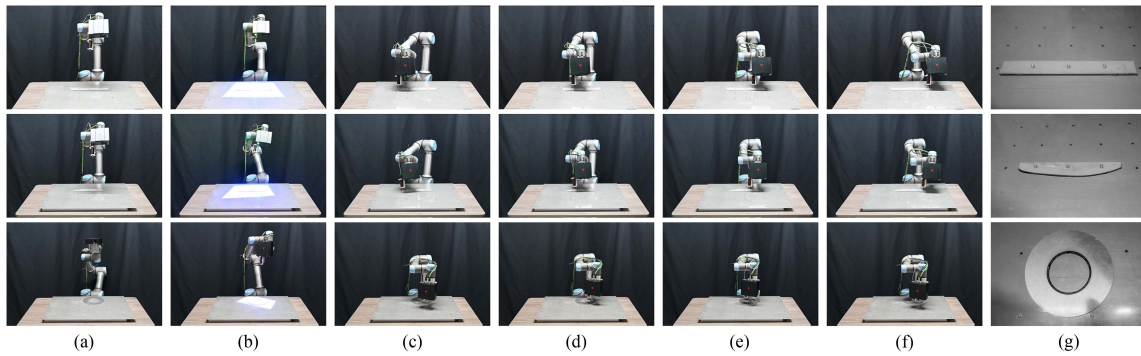


Fig. 13. Physical experiment. (a) Preparing pose. (b) Image capture. (c)–(f) Robotic coating operation. (g) Coated workpieces.

integrating 2-D and 3-D deep learning, our system with LBC achieves higher accuracy with RMSEs of 0.593 (straight), 1.099 (curved), and 1.252 mm (circular). Considering the precision demands of the coating application, these error margins are well within acceptable tolerances.

As shown in Table III, in terms of efficiency, our model maintains a lightweight architecture (7.3 M parameters), ensuring efficient deployment in real-world scenarios. Although the hybrid strategy incurs a moderate computational overhead (483 ms without LBC and 693 ms with LBC)—slightly higher than purely 2-D (357 – 408 ms) or 3-D (409 ms) methods, it significantly outperforms traditional hybrid approaches (1219 ms) in both speed and accuracy. Notably, even without the LBC algorithm, our framework’s accuracy surpasses that of other methods. The decision to employ LBC depends on the desired accuracy–efficiency tradeoff.

Besides, considering the limited capture time of the industrial RGB-D camera (approximately 851 ms in our system), our framework is intended for offline application. Under these conditions, inference times of 483 and 693 ms is highly acceptable. Furthermore, while calculating curvature and concavity during SFRM training takes an average of 5.527 s per frame, the inference time of SFRM is only 61.3 ms. This speedup is attributed to the SFRM’s learned capacity to predict seam features directly.

E. Physical Experiments

To validate the proposed method, physical experiments were conducted using the hardware platform illustrated in Fig. 12. The RGB-D camera (RVC-I2370, RVBUST, Inc.) offers a high repeatability of single point with 0.04 mm and captures RGB-D images at a resolution of 1440×1080 . The 6-DOF robotic

arm (UR5e, Universal Robots, Inc.) has a payload capacity of 5 kg and a repeatability of ± 0.03 mm. Robotic arm control was implemented using the MoveIt! packages within the Robot Operating System (ROS).

The workpieces used in the experiments were conventional aluminum or steel plates with varied seam shapes, representative of those commonly encountered in aviation coating applications. The task involved sealing the seams between the plate workpieces and a baseplate using a glue gun. As shown in Fig. 13, the experimental results robustly demonstrate the system’s capability to accurately identify and effectively seal various types of seams. This validation confirms that the generated glue lines consistently achieve full coverage of the seams, thereby ensuring the integrity and quality of the sealing process across diverse applications.

VII. CONCLUSION

In conclusion, our multimodal detection system effectively enhances robotic seam gluing accuracy by integrating coarse 2-D segmentation with precise 3-D refinement. The dedicated RGB-D coating dataset and ablation studies validate the system’s robustness and superiority. Future directions will focus on improving generalization, multiseam detection, and surface adaptation through simulation, advanced modeling, and hybrid learning strategies.

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